Task 2: The Distribution of the Sample Mean

COURSE NAME

CSDS 413 Introduction to Data Analysis

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Contents

Context

In this task, you will we gain insight about the Central Limit Theorem. Consider three random variables:

(a) A **Uniform distribution** on $[0.25, 1.25]$

(b) A **Power-Law distribution** (with parameter $\alpha = 3$)

(c) A **Pareto distribution** (`np.random.pareto`, with parameter $\alpha = 1$)

1 Part A: Data Generation

- For each distribution, generate **replicates of data** with sizes 10, 100, and 1000.
- Each replicate should consist of 100 samples.
- Example: For "10 replicates," generate 10 separate datasets, each with 100 samples.
- For one replicate from each distribution, plot the distribution of the sample to ensure that it follows a uniform/power-law distribution.

Implementation notes:

- In **Python**, use `numpy.random.uniform` and `numpy.random.power`.
- In **R**, use `runif()`, `rplcon()` (for Power, with shape1 = α, shape2 = 2.5).

2 Part B: Sample Means and Visualization

- Compute the sample means for each set of replicates. You should have 9 of them, 3 for each distribution with varying sample sizes.
- For each distribution and replicate size, we want to view and compare their distributions. Which plot do you think is appropriate? Plot the appropriate plot for each replicate, please plot them separately.
- Do not confuse the distribution of sample mean with the distribution of the sample (which you plotted in Part A).

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3 Part C: Normal Approximation via CLT

- By the Central Limit Theorem, the distributions of sample means should approximate normal distributions. Determine the parameters (mean and standard deviation) of the approximating normal distributions for both power law and uniform cases.
- Overlay the probability density functions (PDFs) of these normal distributions on the corresponding plot from part B.

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4 Part D: Distribution Comparison with KS-Test

- Compare the empirical distributions of sample means (from power law vs. uniform) using the Kolmogorov–Smirnov (KS) test.
- Discuss whether the KS-test supports the theoretical expectation that both sample mean distributions should approximate normality under the CLT. And investigate how the power of the KS-test changes as the number of sample means increases (10, 100, 1000).